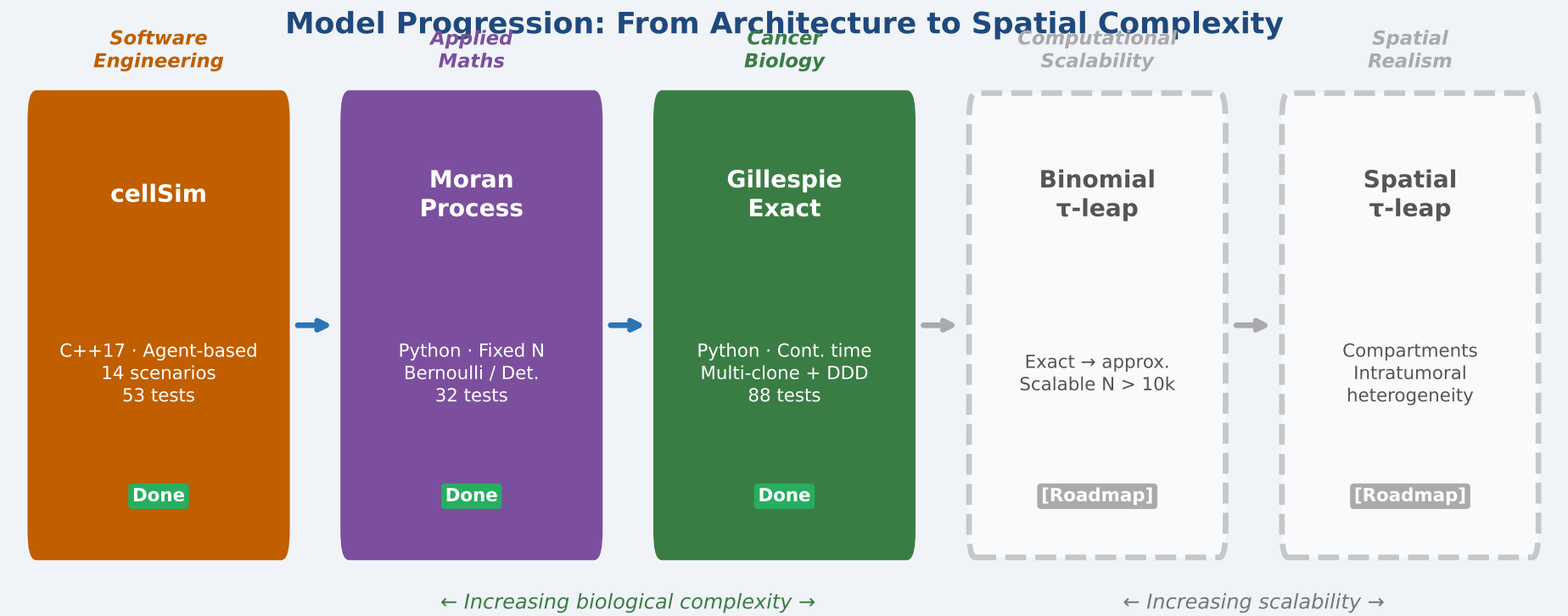
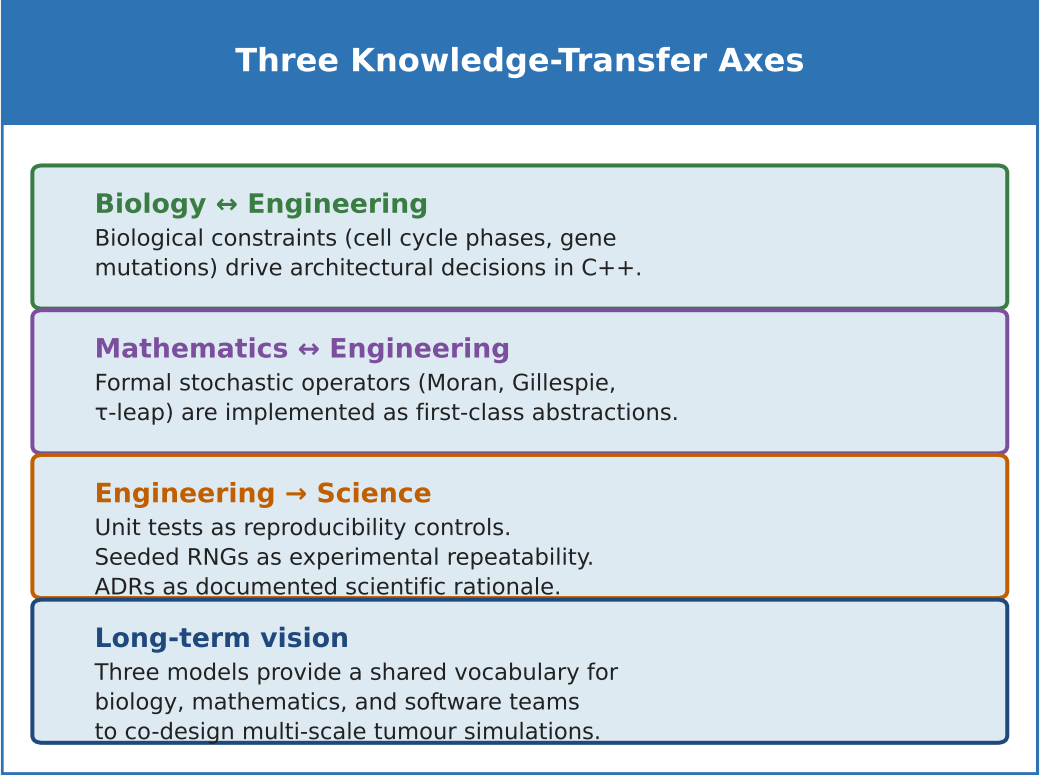
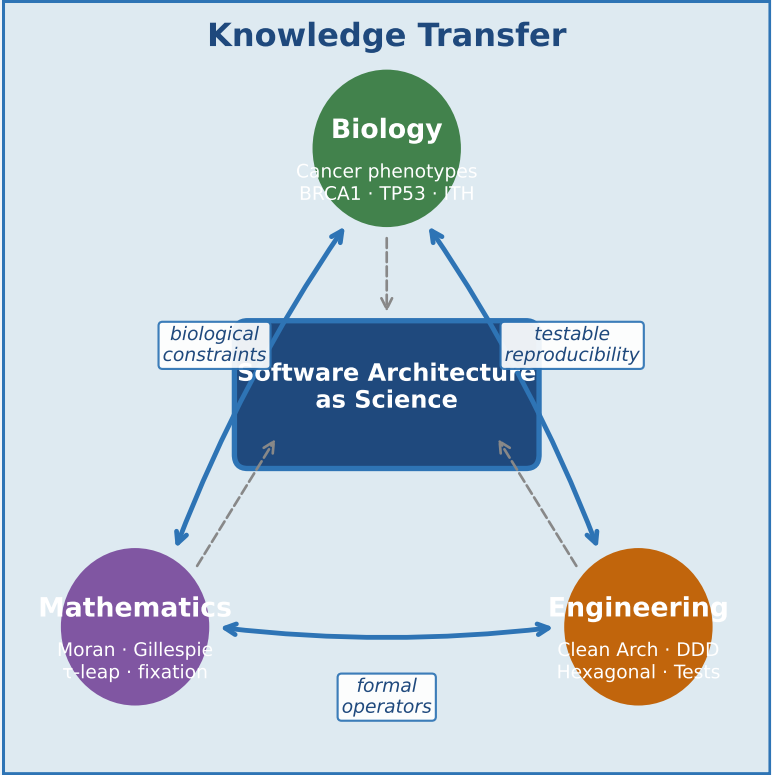


Computational Models of Tumour Cell Evolution

A Multidisciplinary Knowledge-Transfer Framework · April 2026



Architecture as Science	
#	Clean Arch · cellSim C++17 53 tests · 14 validated scenarios
⬡	Hexagonal + DDD · Gillespie 88 tests · domain ≠ infrastructure
↔	Ports & Adapters · Moran 32 tests · strategy = operator
✓	Tests = computational controls ADRs = scientific rationale

Open Questions	
?	cellSim: biological use cases HBOC · tissue types · outputs
?	Gillespie: calibrate vs. TCGA longitudinal sequencing data
↔	Cross-model scenario comparison cellSim ↔ Gillespie
⌚	Binomial τ -leap: step-size strategy for tumour N > 10k

Roadmap & Grant Framing	
[1]	Phase 1 complete: 3 models 173 tests · open-source · reproducible
[2]	Phase 2: Binomial τ -leap scalability to clinical N
[3]	Phase 3: Spatial τ -leap intratumoral heterogeneity (ITH)
[P]	Manuscript in preparation SoftwareX · PLOS Comp. Biol.